

31. Ch 16, Kernel Methods

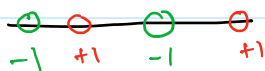
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Refs: ch. 16 Shalev-Schwartz

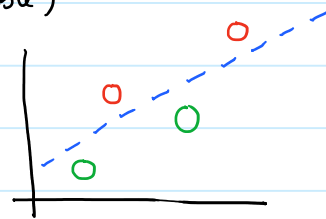
ch. 2 Fasshauer and McCort 2015
"Kernel-based approximation
methods using Matlab"

A common trick is to embed data x into a higher dimension $\Psi(x)$, especially because if the embedding Ψ is nonlinear, then a linear classifier in high-dimensional space corresponds to a non-linear classifier in the original space (but the non-linearity is done in pre-processing, conceptually at least, so it doesn't hurt things like convexity)

Example 1D points (not linearly separable)



embed into
2D via
 $\Psi: \mathbb{R}^1 \rightarrow \mathbb{R}^2$



... now in 2D, the dataset is linearly separable

putting in #'s to make a real example:

Consider a dataset w/ "x" as $\{-10, -9, \dots, 9, 10\}$

let labels y be defined as $y = \begin{cases} +1 & |x| > 2 \\ -1 & |x| \leq 2 \end{cases}$

So not lin. separable

but define $\Psi: \mathbb{R}^1 \rightarrow \mathbb{R}^2$ as $\Psi(x) = (x, x^2)$, and
now it is lin. sep. in $\mathbb{R}^2$



So, in theory, it's simple:

$\Psi: X \rightarrow F$ ← new feature space,

preprocess data. Our distribution D over $X \times Y$ is now extended to D^Ψ over $F \times Y$
in the obvious way, so $L_{D^\Psi}(h) = L_D(h \circ \Psi)$

1965 Cover's Thm: (paraphrasing) embedding n points into a higher dimension is more likely to make them linearly separable

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we've even seen this trick before: we never discussed **quadratic** or **polynomial** regression because we can treat them as a special case of **linear** regression (after adding extra features)

For most learning algorithms to work, we'll need F to at least be a **Hilbert space**. $\leftarrow$ inner prod. space that's "complete" Ex: $L^2(\mathcal{X})$
Like $\mathbb{R}^n$ but: $\cdot$ not all vectors allowed $L^2(\mathbb{R}) \dots$
 $\cdot$ bases are different
 $\cdot$ norms not equiv., different topologies...

The kernel trick

Often it's unclear what Ψ to use, and/or we don't want to work in F (why? it's larger-dim, even ∞ -dim!)

The kernel trick is a way around this, applicable to many (not all) algorithms.

Subtlety: w defines a $h \in \mathcal{H}$ via:

Ex Kernel Ridge Regression (Saunders et al., ICML '98) $h(x) = \langle w, x \rangle$
(my simplified derivation w/o duality)

$$\min_w \frac{1}{2} \|Xw - y\|^2 + \lambda \|w\|^2$$

$$\begin{matrix} \leftarrow d \\ \uparrow m \\ \downarrow \end{matrix} \begin{bmatrix} X \\ \hline \end{bmatrix} \xrightarrow{\quad} x_i^T$$

Solution is $w = (X^T X + \lambda I)^{-1} \cdot X^T y$

$$b = \sum_{i=1}^m y_i x_i$$

Now, suppose X is

$$f_i = \psi(x_i) \in F \quad \begin{matrix} \infty? \\ \begin{bmatrix} \vdots \\ \psi(x_i)^* \\ \vdots \end{bmatrix} \end{matrix} \quad \text{or } f_i^*$$

our sol'n is still valid in any Hilbert space, but

$$\min_{w \in F} \frac{1}{2} \sum_{i=1}^m (\langle \psi(x_i), w \rangle_F - y_i)^2 + \lambda/2 \|w\|_F^2$$

$$X^T X = \sum_{i=1}^m \psi(x_i) \psi(x_i)^* \quad \infty \times \infty$$

isn't easy to compute

$$\langle f, g \rangle_F = \int f(t)g(t)dt \quad \text{or whatever}$$

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Trick:

$$(X^T X + \lambda I)^{-1} = I - X^T (I + X X^T)^{-1} X$$

(matrix inversion lemma)
aka Sherman-Morrison-Woodbury

From here on, choose $\lambda = 1$ for simplicity

true in finite dim., also true in ∞ dim. with perhaps a few caveats we won't worry about

$\rightarrow$ a $\tilde{d} \times \tilde{d}$ matrix if $\psi: \mathbb{R}^d \rightarrow \mathbb{R}^{\tilde{d}}$

How does that help? $X X^T = \sum_i \psi(x_i) \psi(x_i)^T$ is tricky

$$(X X^T)_{ij} = \psi(x_i)^T \psi(x_j) = \langle \psi(x_i), \psi(x_j) \rangle$$

$k(x, \tilde{x}) = \langle \psi(x), \psi(\tilde{x}) \rangle$ is a kernel

So, ψ defines K , but...

you can define K without needing to know what ψ is!

! we often refer to k and K both as the "kernel"

In particular, let the kernel matrix be K such that

aka Gram matrix

$$(K)_{ij} = k(x_i, x_j) = \langle \psi(x_i), \psi(x_j) \rangle, \quad K \in \mathbb{R}^{m \times m}$$

So, back to ridge regression, $\lambda = 1$ still

$$w = (I - X^T (I + K)^{-1} X) \cdot b, \quad b = X^T y = \sum_i y_i \psi(x_i)$$

$$= b - X^T \tilde{K} X b, \quad \tilde{K} = (I + K)^{-1} \in \mathbb{R}^{m \times m}$$

↑
w/o knowing ψ ,
this isn't computable

$w \in F, b \in F$
by themselves,
hard to represent...
but $\langle w, b \rangle$ is ok!

but... we don't actually want w , we want to apply h , i.e. $h(x)$

So, can we compute $h(x) = \langle w, x \rangle$?

or, since we're embedding x to $\psi(x)$, can we compute $h(x) = \langle w, \psi(x) \rangle$?

yes! $w = b - X^T \tilde{K} X b$

$$\begin{aligned} \langle b, \psi(x) \rangle_F &= \langle \sum_i y_i \psi(x_i), \psi(x) \rangle_F \\ &= \sum_i y_i \langle \psi(x_i), \psi(x) \rangle_F = \sum_i y_i \underline{k(x_i, x)} \end{aligned}$$

computable

$$\text{and } \langle X^T \tilde{K} X b, \psi(x) \rangle_F = \langle \psi(x), X^T \tilde{K} X^T b \rangle_F$$

$$= c^T \tilde{K} d \quad \text{where} \quad c_i = \langle \psi(x), \psi(x_i) \rangle$$

$$d_i = \langle \psi(x_i), \sum_j y_j \psi(x_j) \rangle$$

So all computable via only

knowing $K(x, \tilde{x})$ (ψ is completely implicit)!

That's the kernel trick

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General theory

The kernel trick applied to ridge regression.

Does it apply to SVM? (yes) Boser et al. '92

What else?

Thm 16.1 "Representer Thm" Schölkopf et al. 2000

Suppose $\Psi: X \rightarrow \mathcal{F}$ — Hilbert space now, not hypothesis class

and let $\phi \neq \argmin_{w \in \mathcal{F}} f(\langle w, \Psi(x_1) \rangle, \dots, \langle w, \Psi(x_m) \rangle) + R(\|w\|_{\mathcal{F}})$ Hilbert space norm

for arbitrary f and R is nondecreasing on $\mathbb{R}_+$

(ex: $f(a_1, \dots, a_m) = \frac{1}{m} \sum_i \ell(a_i)$, $R(a) = a^2$)

Note Hard + Soft SVM, and Ridge Regression, satisfy the assumptions

then $\exists \alpha \in \mathbb{R}^m$ s.t.

$w = \sum_i \alpha_i \Psi(x_i)$ (ie, $w \in \text{span}\{\Psi(x_1), \dots, \Psi(x_m)\}$)
is optimal. and we can use the kernel trick,

since now $\langle w, \Psi(x) \rangle$ is computable

Proof:

$w^* \in \mathcal{F}$

let w^* be an optimal sol'n, decompose it as

$$w^* = w + u, \quad w \in \text{span}\{\Psi(x_1), \dots, \Psi(x_m)\} = V \subseteq \mathcal{F}$$

So via orthogonality,

$$u \in V^\perp \quad \text{since } \mathcal{H} = V \oplus V^\perp \quad (\dim(V) < \infty)$$

(so $V = \bar{V}$, thus $\mathcal{H} = \bar{V} \oplus V^\perp = V \oplus V^\perp$)

$$\|w^*\|^2 = \|w\|^2 + \|u\|^2 \Rightarrow \|w\| \leq \|w^*\|$$

$$\text{and } R \text{ nondecreasing} \Rightarrow R(\|w\|) \leq R(\|w^*\|)$$

Since w^* and w differ only on $V^\perp$,

they both have the same value of $f(\langle w, \Psi(x_i) \rangle, \dots, \langle w, \Psi(x_m) \rangle)$

hence if w^* is optimal, w must be optimal

as well, and $w \in V$ as desired. $\square$

(book has slight error: it didn't assume optimal solution exists)

Conclusion: Kernel-SVM is practical

→ either use dual problem,
or primal problem using $\phi(x_i)$ as the basis

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Examples of kernels

Given $K(\cdot, \cdot)$, Sometimes we have an explicit expression for ψ , but not always

(and not just any $K(\cdot, \cdot)$ works ... more on that later)

① Polynomial Kernel, $c > 0$ See §3.4.1 Fasshauer "dot product kernel"

$$K(x, y) = (c + \langle x, y \rangle)^p \quad (\psi \text{ not unique})$$

$$= \langle \psi(x), \psi(y) \rangle \text{ for } \psi: \mathbb{R}^d \rightarrow \mathbb{R}^{\tilde{d}}, \quad \tilde{d} = \binom{d+p}{p} \text{ or } \tilde{d} = (d+1)^p \text{ depends on exact embedding}$$

$$\text{ex, } d=2, p=2, \tilde{d}=6$$

$$K(x, y) = (c + x_1 y_1 + x_2 y_2)^2 = \begin{bmatrix} x_1^2 \\ x_2^2 \\ \sqrt{2} x_1 x_2 \\ \sqrt{2} c x_1 \\ \sqrt{2} c x_2 \\ c \end{bmatrix}^T \begin{bmatrix} y_1^2 \\ y_2^2 \\ \sqrt{2} y_1 y_2 \\ \sqrt{2} c y_1 \\ \sqrt{2} c y_2 \\ c \end{bmatrix}$$

For larger d , you could form ψ explicitly but it's not a good idea.

Formula for K costs $O(d)$ per pair x, y
vs. $O(\tilde{d})$ using embedding explicitly

② Gaussian Kernel (aka RBF, Radial Basis Fun, or "SE")

Most common in practice

$$K(x, x') = e^{-\frac{\|x - x'\|^2}{2\sigma^2}} \leftarrow \text{for } \sigma > 0 \dots \text{or often we write } e^{-\frac{\|x - x'\|^2}{2\sigma^2}} \text{ Squared Exponential}$$

what is ψ ?

consider $d=1$ case,

$$K(x, x') = e^{-\frac{(x-x')^2}{2}} \quad (\sigma=1 \text{ for now})$$

define $\psi: \mathbb{R} \rightarrow \ell^2$ by ℓ^2 on $\{0, 1, 2, \dots\}$

$$(\psi(x))_n = \frac{1}{\sqrt{n!}} e^{-x^2/2} x^n$$

$$\text{so } \|\psi(x)\|_{\ell^2}^2 = e^{-x^2} \cdot \left(\sum \frac{1}{n!} (x^2)^n \right)$$

$$= 1 \quad \text{since } e^a = \sum \frac{1}{n!} a^n \text{ has } \infty \text{ radius of convergence}$$

and note

$$\langle \psi(x), \psi(x') \rangle_{\ell^2} = \sum (\dots) \quad (\dots)$$

$$= \exp\left(-\frac{x^2}{2} - \frac{x'^2}{2}\right) \sum_{n=0}^{\infty} \frac{(xx')^n}{n!} = e^{-\frac{x^2}{2} - \frac{x'^2}{2} + xx'}$$

$$= \exp\left(-\frac{x^2}{2} + xx' - \frac{x'^2}{2}\right)$$

$$= \exp\left(-\frac{(x-x')^2}{2}\right)$$

Exercise '26: given ψ , deduce K

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③ Sigmoid Kernel

$$a, b \geq 0, \quad K(x, x') = \tanh(a \langle x, x' \rangle + b)$$

(w/ SVM, this is similar to a simple neural network)

④ Matern kernel

Discussion

The choice of kernel is one way to encode prior knowledge (just as choice of features was)

Similar to classic statistics: for regression,

if x_i is a covariate, sometimes you

choose to use $\log |x_i|$ instead

Often polynomial or Gaussian kernels are the only choices considered (and their parameters are hyperparameters that are fit)

... but kernels also used many places

(ex: Gaussian processes, Kriging in spatial stat.)

and there, often custom kernels are used.

For more ex., see exercises in ch 6 in Mohri et al. (4 Fasshauer)

Often do MLE to fit hyperparameters of the kernel

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What are the restrictions on K ? ① Symmetry

② ^{Shalev-Schwartz} Lemma 16.2 (Mercer's Thm, ^{very} simplified) ^{This is combining several thms}

A symmetric function $K: X \times X \rightarrow \mathbb{R}$ implements an inner product in some Hilbert space iff K is psd (pos semidef.)
^{check via Bochner}

A kernel is psd if $\forall m, \forall \{x_1, \dots, x_m\}$, the Gram matrix

$G_{ij} = K(x_i, x_j)$ is a psd matrix

$$G = G^T, G \succeq 0 \text{ iff } \lambda_i(G) \geq 0 \quad \forall_i$$

There are also other equivalent definitions, see a more comprehensive ref. like Fasshauer.

The book says "psd" but that's a typo. They mean "positive definite" (and $\lambda_i(G) > 0$ not ≥ 0)

Turns out not all Hilbert spaces can be described w/ a kernel

Those that can are **Reproducing Kernel Hilbert Spaces (RKHS)**

In particular, a RKHS has the property that it is over an underlying domain Ω (think of $L^2(\Omega)$)

and if $x \in \Omega$,

$f \in \mathcal{H}$, then $\delta_x \in \mathcal{H}^* (= \mathcal{H})$

for $\Omega = [0, 1]$

or $\Omega = \mathbb{R}$,

or $L^2(\Omega)$ for $\Omega = \mathbb{N}$ or $\mathbb{Z}$

Using $\mathcal{H}$ to denote generic Hilbert Space now, not a set of classifiers

When $\Omega \subseteq X$ we can think of $x \in \Omega$ as a datapoint.

So there's a canonical embedding $\Psi \dots$

ex: $\mathcal{H} = L^2(\mathbb{N})$, $\Omega = \mathbb{N}$,

$f \in L^2$ is $f = (f_1, f_2, f_3, \dots)$

then $\delta_n(f) = f_n$ is a bounded linear functional

$L^2(\mathbb{N})$ is a RKHS (so is $L^2(\mathbb{Z})$)

ex: $\mathcal{H} = L^2(\Omega = [0, 1])$

$f \in L^2$, $x \in [0, 1]$,

$\delta_x(f) = f(x)$ ($= \int f(x') \delta(x' - x) dx'$)
^{in physics notation}

this is not bounded

(not even well-defined, since L^2 is not defined pointwise)

So $L^2([0, 1])$ is not a RKHS (nor is $L^2(\mathbb{R})$)

More precise: Thm 2.4 Fasshauer

Let $\mathcal{H}$ be a Hilbert space of real-valued functions over $\Omega \subseteq \mathbb{R}^d$ w/ inner product $\langle \cdot, \cdot \rangle$. Then $K: \Omega \times \Omega \rightarrow \mathbb{R}$ is a reproducing kernel for $\mathcal{H}$ if

1) $K(\cdot, x) \in \mathcal{H} \quad \forall x \in \Omega$

2) $\langle f, K(\cdot, x) \rangle = f(x) \quad \forall f \in \mathcal{H} \quad \forall x \in \Omega$

i.e. $K(\cdot, x) = \delta_x$

See Fasshauer p. 34 + ch. 8

... but **Sobolev spaces** are RKHS.

$H^K(\mathbb{R}^d) =$ functions w/ K square-integrable weak derivatives, w/ appropriate inner product

(If $K > d/2$ then $H^K \subseteq C$)

Their kernel is the **Matérn kernel**

RKHS are like generalized Sobolev spaces

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History

Kernels in general, Mercer (1909)
Schoenberg (1938)
* Aronszajn (1950)

In ML, Boser, Guyon, Vapnik '92 (SVM)
Schölkopf et al '98 (general)

Supplement

Forming the Gram matrix is slow!
 $O(m^2 d)$ computation

$$m \begin{bmatrix} K \end{bmatrix} \quad K_{ij} = K(\underbrace{x_i, x_j}_{\in \mathbb{R}^d})$$

many attempts to improve

One broad strategy is the Nyström method
(also used in integral eq'n) which
picks out representative pts.

A well-known 2007 paper (Rahimi, Recht) "Random Fourier Features"
makes approx. maps ψ st.

$$K(x, x') \approx \langle \psi(x), \psi(x') \rangle$$

for **Shift-invariant kernels**

$$K(x, x') = G(x - x') \quad \underline{\text{Ex}}$$

$$\text{Gaussian} \quad \exp(-\frac{1}{2} \|x - x'\|^2)$$

$$\text{Laplacian} \quad \exp(-\|x - x'\|)$$

$$\text{Cauchy} \quad \prod_{i=1}^d \frac{1}{1 + (x_i - x'_i)^2}$$

relies on...

Bochner's Thm (see next pg.)

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Warmup A $n \times n$ matrix K is positive definite if $\forall x \in \mathbb{R}^n, x \neq 0$,
then $x^T K x > 0$. Theorem: If K is symmetric*, it is
positive definite iff all its eigenvalues are strictly positive.

Also note: if the kernel is shift-invariant,
then it is diagonalized by the FOURIER TRANSFORM
(if periodic or so ... basically, neglect boundaries)

Proper statement:
DFT diagonalizes
circulant matrices

* if not symmetric,
this isn't true.
 $K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ has eigs $\{1, 1\}$
but not pos. def.
since its symmetric part,
 $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, is singular.

Bochner's Thm A cts kernel of the form $K(x, x') = G(x - x')$ defined
over a locally compact set X is pos. def. iff G is the
Fourier transform of a non-neg. measure, i.e., $G(x) = \int_X e^{i\omega x} \underbrace{p(\omega) d\omega}_{\text{like eigenvalues}}$ ^{like eigenfunctions}

For most common shift-invariant kernels, $p(\omega)$ is known

So, the random Fourier Features idea:

approximate $G(x)$ via Monte Carlo integration: draw a set

$\{\omega_j\}_{j=1}^N$, iid, $\omega_j \sim p(\omega)$, equivalent to an $G(x) \approx \frac{1}{N} \sum_{j=1}^N e^{i\omega_j x}$
 $\omega_j \sim p(\omega)$

approx. feature map into $\mathbb{R}^N$. Of course MC converges very slowly
but not too bad if X has small covering number

(and in practice, low-accuracy is OK, since we don't need to approximate
the kernel very well, just enough to give us "features" to learn from)

See Martinsson, Tropp 2020 survey

Other kernel methods

- Probably 3 most common kernel methods
- (1) Kernel-SVM
 - (2) Kernel-ridge regression
 - (3) Kernel-PCA $\rightarrow$ see §14.4.4 in Kevin Murphy's text (trick for centering data)
- $\rightarrow$ or more generally, inside a Generalized Linear Model (GLM)
cf. §14.3 Murphy

(4) Kernelized nearest-neighbor and K-medoids

See §14.6 Murphy for comparison of kernel methods

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"Kernel" is used in other contexts too, eg,

1D Kernel $K(x)$ st. $\int K(x)dx = 1$, $\int x K(x)dx = 0$,

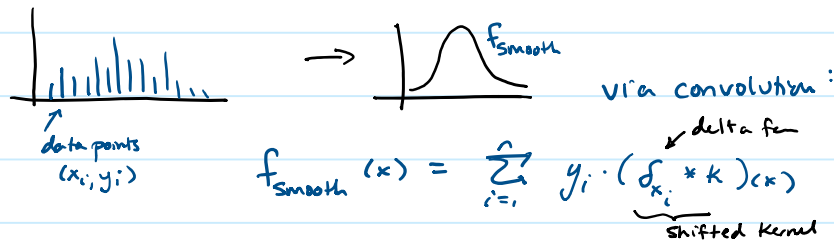
like Gaussian Kernel

$$\int x^2 K(x)dx > 0$$

interpolation
↓
"Scattered data
fitting"

used for kernel density estimation or Parzen window dens. est.

Idea is to turn an empirical histogram into an estimate of the pdf



where convolution is

$$(f * K)(x) := \int f(y) K(x-y) dy.$$

Other examples of kernels

↗ Gaussian Processes, see next lecture

Matern kernel, common in GP, **Kriging**

based on modified Bessel function

(shift invariant)

String kernels for natural language processing (NLP)

See **Murphy** for more examples (or Fasshauer)

"Machine Learning"